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Human vs. Machine Minds: Ego-Centric Action Recognition Compared

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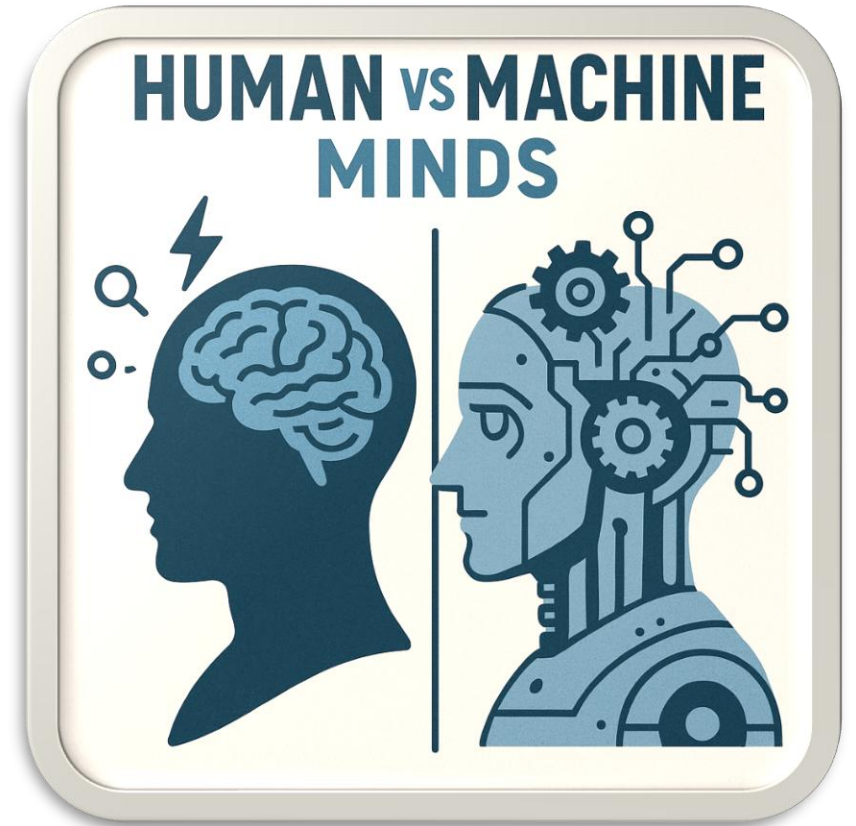


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Motivation

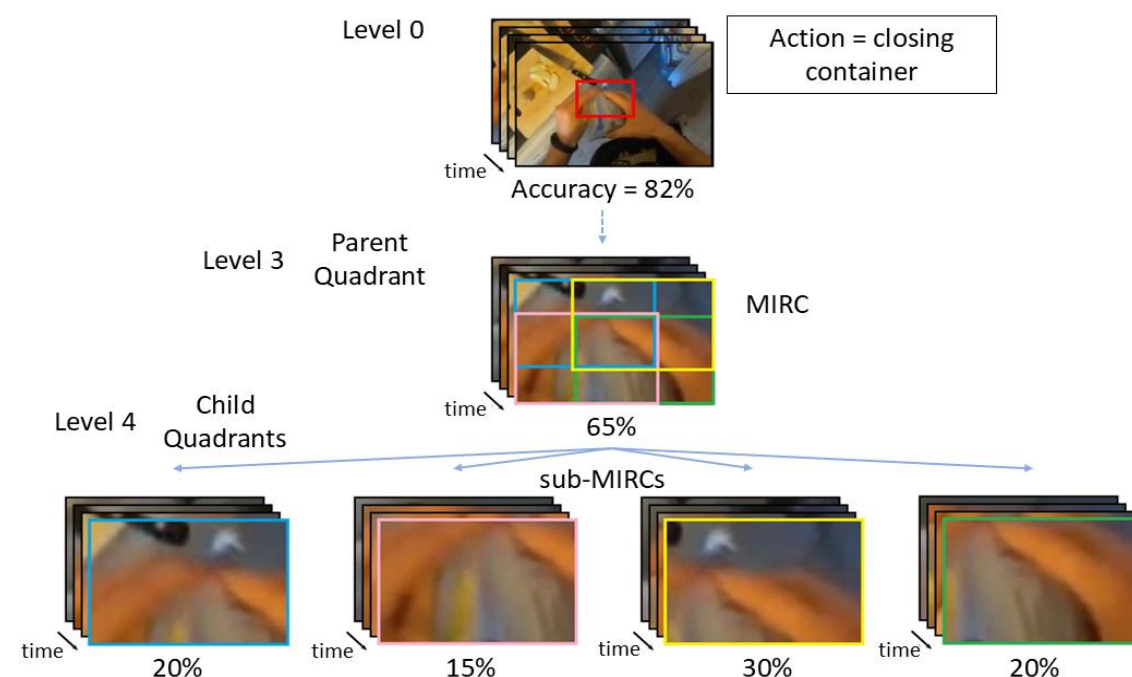
- Is the Recognition of action different between humans and AI models?
 - ✓ We test their performance in challenging real-world ego-centric scenarios involving occlusion, clutter, and low-resolution.
 - ✓ This study seeks to understand *how* and *why* human and machine recognition diverge by introducing a new benchmark, **Epic ReduAct**



Epic ReduAct dataset

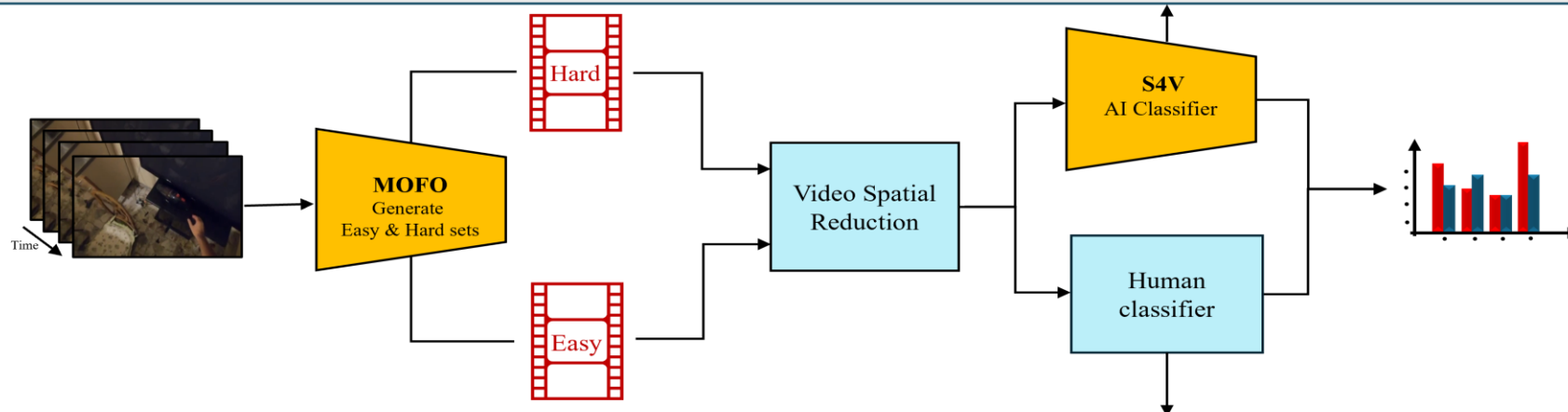
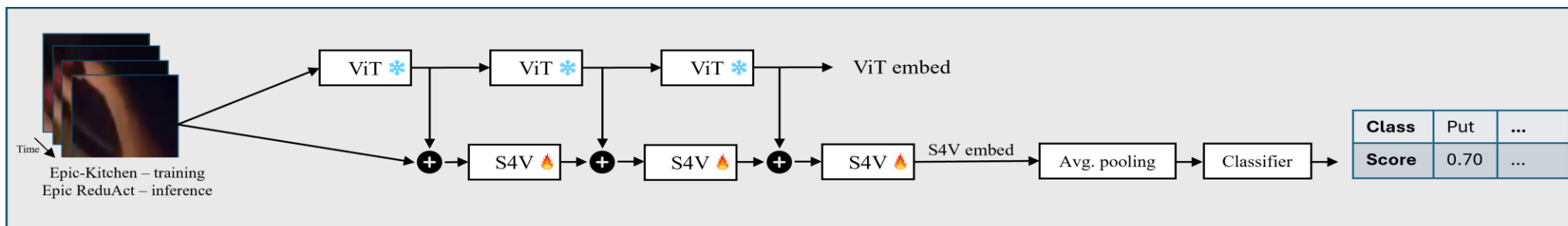


- What is Epic ReduAct?
 - ✓ A systematically reduced version of videos from the **Epic-Kitchens-100** dataset, designed to study **minimal visual requirements** for recognizing ego-centric human actions.
 - ✓ Identify the smallest recognisable quadrants or **Minimal Recognisable Configurations (MIRCs)**
 - ✓ Actions span common kitchen tasks (e.g., cut, pour, wash)
 - ✓ Created two sets; Easy/Hard, based on AI model (MOFO) performance
 - ✓ **We have reduced the videos to find the MIRCs, using that as a basis for human understanding, and then comparing it with AI**

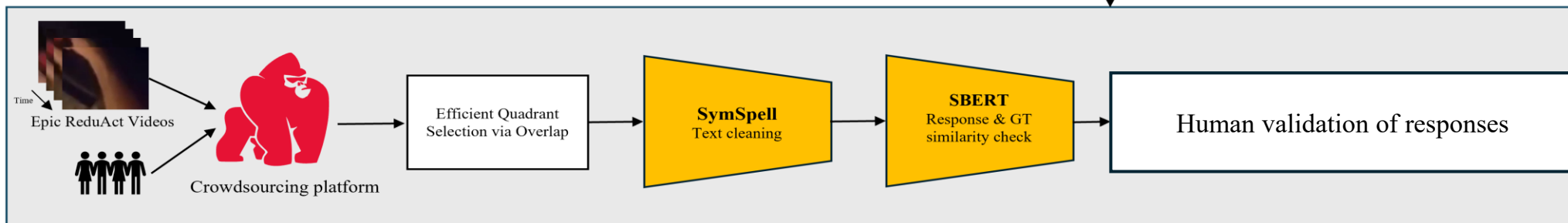


Methodology & Pipeline

AI Classifier

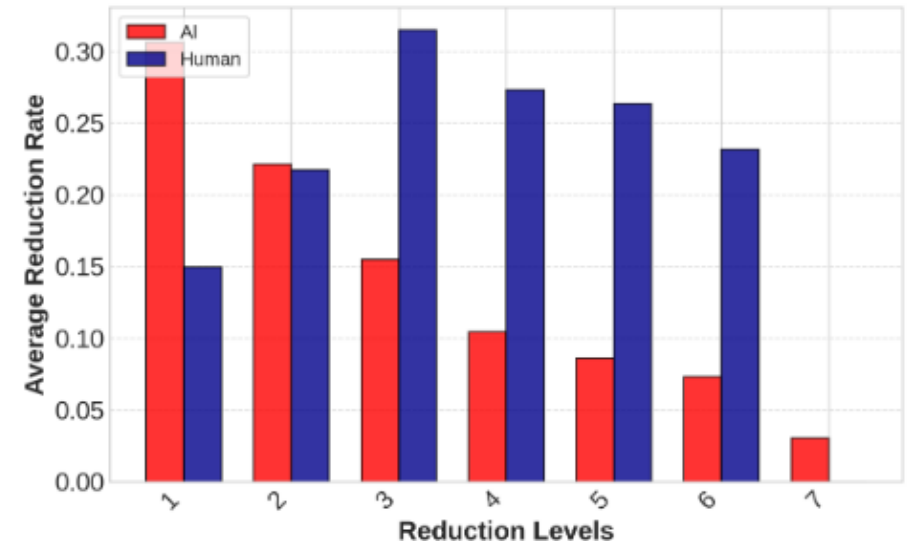


Human Classifier

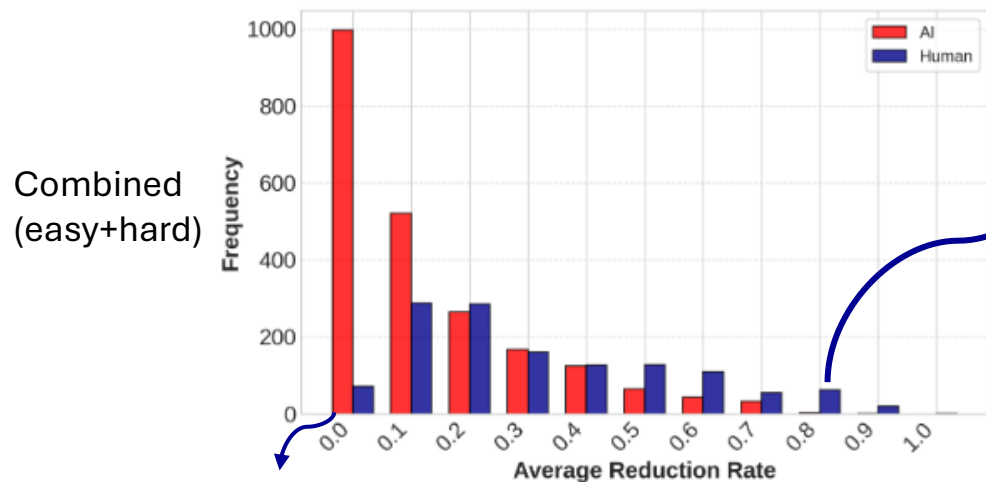


Experiments & Results - *Average Reduction Rate*

- **Metric:**
 - ✓ Average reduction rate as a function of reduction level for all parent-child quadrant pairs.
- **AI Model Performance:**
 - ✓ Accuracy declines gradually as spatial reduction increases.
- **Human Performance:**
 - ✓ Recognition stable at initial levels of spatial reduction.
 - ✓ Drops sharply at higher levels of reduction.



Easy set



Combined
(easy+hard)

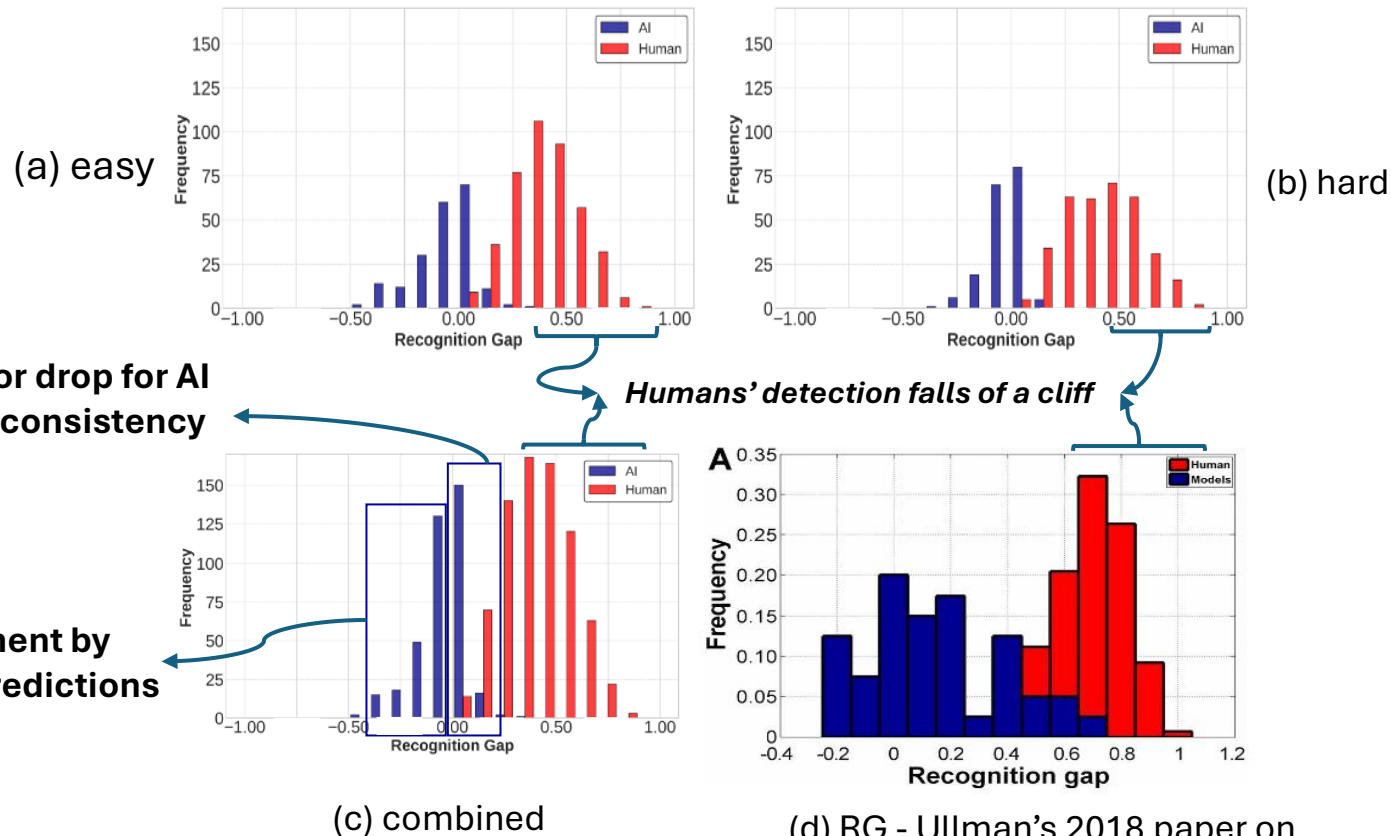
~ 100 videos with a high drop

~ 1000 videos with the lowest drop

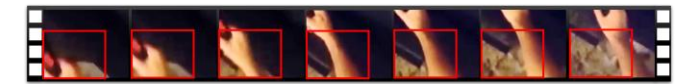
- **Metric:** Distribution of pairs at all reduction levels.
- **Pattern:** Humans' recognition drops significantly while AI model remains more robust in recognition

Experiments & Results - *Recognition Gap*

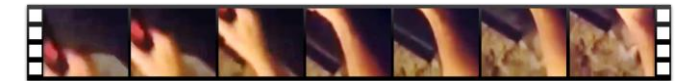
- The difference in average classification accuracy between full MIRC and their sub-MIRC quadrants. (for measuring the effects of spatial feature)
- **Metric:** Frequency distribution of recognition gap.



a) Original video – No reduction – GT put



b) Level 2 – The MIRC level

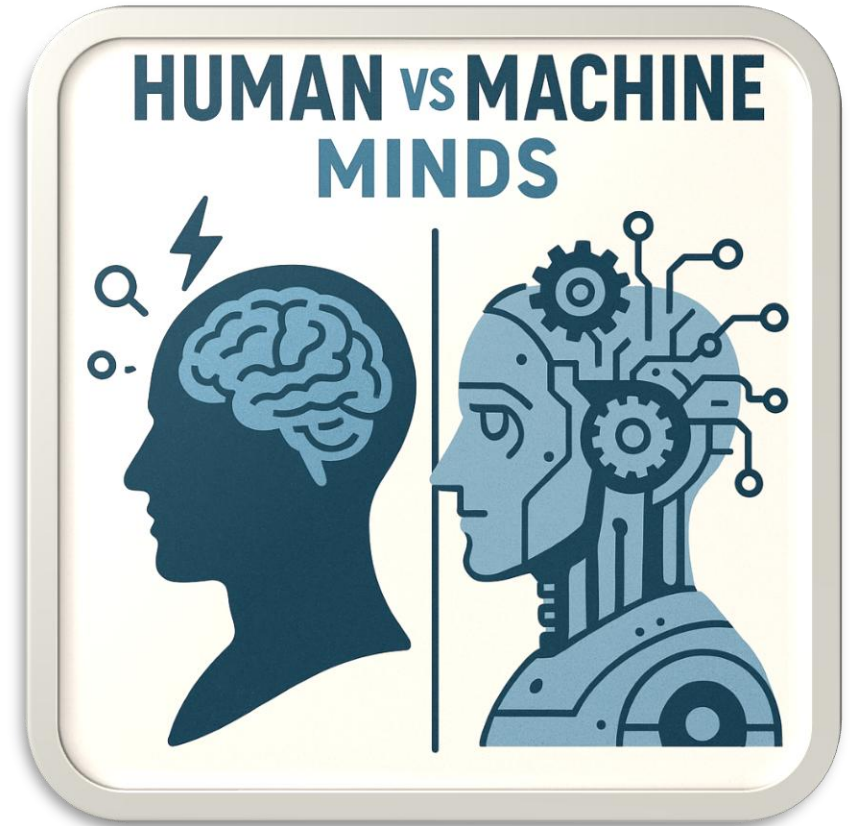


c) Level 3 – The sub-MIRC level of Fig. b

(d) RG - Ullman's 2018 paper on image-based recognition

Conclusion and future works

- **What is the takeaway?**
 - ✓ Humans excel in complex recognition but fail with minimal input, while AI stays more robust.
 - ✓ Future work will explore temporal cues and multimodal data to better align AI with human perception.





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